

## KRUGER MPUMALANGA INTL, South Africa

WMO: 682904

Lat: 25.388S

Lon: 31.099E

Elev: 2829

StdP: 13.25

Time Zone: 2.00 (E02)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 44.9       | 47.8 | 15.9                          | 13.6 | 68.5 | 20.8 | 17.3 | 67.3 | 16.2                  | 65.6 | 14.2 | 65.2 | 4.1                   | 260  | 0.335 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 15.6                   | 91.4            | 64.8 | 88.1 | 64.7 | 86.0 | 64.8 | 72.7                | 81.9 | 71.5 | 80.8 | 70.5 | 79.0 | 9.8                  | 60   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 70.1                            | 123.2 | 75.7 | 68.7 | 117.3 | 74.5 | 68.0 | 114.5 | 73.9 | 38.4          | 81.6 | 37.3 | 81.4 | 36.2 | 79.2 | 90.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 15.3          | 13.7 | 12.2 | DB                         | 40.3 | 97.5               | 2.3 | 3.5                                                | 38.6 | 100.0      | 37.3 | 102.0      | 36.0 | 103.9      | 34.2 | 106.4 | (4) |
| (5)               |      |      | WB                         | 34.8 | 74.7               | 1.6 | 1.5                                                | 33.6 | 75.7       | 32.7 | 76.6       | 31.8 | 77.4       | 30.7 | 78.4  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 67.3        | 72.3  | 72.6  | 71.4  | 67.4  | 64.7  | 60.7  | 59.8  | 62.8  | 67.1  | 68.1  | 69.4  | 71.5  | (6)  |  |
|                                                                        | DBStd       | 6.42        | 3.63  | 3.17  | 3.62  | 4.29  | 4.03  | 4.08  | 4.47  | 5.26  | 6.34  | 6.28  | 5.73  | 5.17  | (7)  |  |
|                                                                        | HDD50       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD65       | 626         | 1     | 1     | 3     | 26    | 54    | 139   | 173   | 107   | 52    | 37    | 24    | 9     | (9)  |  |
|                                                                        | CDD50       | 6310        | 692   | 632   | 663   | 521   | 456   | 322   | 303   | 398   | 514   | 560   | 581   | 668   | (10) |  |
|                                                                        | CDD65       | 1461        | 228   | 213   | 201   | 97    | 45    | 11    | 11    | 40    | 116   | 132   | 155   | 212   | (11) |  |
|                                                                        | CDH74       | 9957        | 1228  | 1090  | 1078  | 484   | 320   | 143   | 143   | 460   | 1218  | 1231  | 1143  | 1419  | (12) |  |
|                                                                        | CDH80       | 2840        | 300   | 234   | 253   | 86    | 49    | 14    | 18    | 116   | 449   | 484   | 390   | 447   | (13) |  |
| (14) Wind                                                              | WSAvg       | 5.6         | 5.9   | 5.7   | 5.1   | 4.8   | 4.6   | 5.0   | 5.1   | 5.9   | 6.6   | 6.5   | 6.2   | 5.9   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 31.7        | 5.5   | 5.2   | 4.3   | 2.0   | 0.6   | 0.4   | 0.3   | 0.4   | 0.7   | 2.4   | 4.8   | 5.0   | (15) |  |
|                                                                        | PrecMax     | 60.5        | 11.1  | 23.9  | 8.5   | 4.3   | 2.8   | 1.8   | 1.9   | 1.2   | 3.3   | 4.3   | 8.5   | 9.3   | (16) |  |
|                                                                        | PrecMin     | 14.1        | 0.7   | 1.5   | 1.4   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.8   | 1.2   | 1.2   | (17) |  |
|                                                                        | PrecStd     | 10.7        | 2.8   | 5.6   | 2.2   | 0.9   | 0.6   | 0.4   | 0.4   | 0.4   | 0.8   | 1.1   | 1.7   | 1.9   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 89.9  | 89.5  | 89.4  | 86.2  | 85.6  | 82.2  | 82.6  | 87.9  | 93.2  | 95.2  | 93.3  | 93.3  | (19) |  |
|                                                                        |             | MCWB        | 68.7  | 69.4  | 68.0  | 66.1  | 58.5  | 55.5  | 54.1  | 57.8  | 59.0  | 62.8  | 65.1  | 69.9  | (20) |  |
|                                                                        | 2%          | DB          | 86.0  | 85.8  | 85.8  | 82.4  | 80.7  | 78.4  | 78.6  | 84.0  | 89.3  | 89.8  | 88.2  | 89.3  | (21) |  |
|                                                                        |             | MCWB        | 68.3  | 69.0  | 66.7  | 64.6  | 59.0  | 54.1  | 53.2  | 57.3  | 59.1  | 63.4  | 66.2  | 69.4  | (22) |  |
|                                                                        | 5%          | DB          | 82.8  | 82.6  | 82.7  | 79.1  | 77.2  | 74.9  | 73.8  | 79.2  | 85.8  | 84.6  | 84.2  | 84.6  | (23) |  |
|                                                                        |             | MCWB        | 68.8  | 68.3  | 66.9  | 64.6  | 58.7  | 53.3  | 52.5  | 55.7  | 58.7  | 62.5  | 66.0  | 69.2  | (24) |  |
|                                                                        | 10%         | DB          | 80.6  | 80.6  | 80.4  | 76.7  | 74.6  | 71.4  | 70.1  | 75.4  | 82.0  | 80.7  | 80.6  | 82.0  | (25) |  |
|                                                                        |             | MCWB        | 68.2  | 67.9  | 66.9  | 63.9  | 58.1  | 53.5  | 52.5  | 54.9  | 58.6  | 62.1  | 66.1  | 68.3  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 75.5  | 73.6  | 72.9  | 71.0  | 66.3  | 59.3  | 58.9  | 62.8  | 65.7  | 69.7  | 71.9  | 73.7  | (27) |  |
|                                                                        |             | MCDB        | 83.7  | 83.3  | 80.8  | 77.5  | 76.4  | 70.8  | 70.9  | 75.2  | 79.5  | 85.3  | 82.1  | 84.9  | (28) |  |
|                                                                        | 2%          | WB          | 72.9  | 71.6  | 71.0  | 69.0  | 63.7  | 57.8  | 57.0  | 60.8  | 63.6  | 66.7  | 70.2  | 71.9  | (29) |  |
|                                                                        |             | MCDB        | 80.7  | 81.7  | 79.8  | 75.9  | 72.0  | 68.3  | 66.9  | 73.8  | 76.5  | 79.0  | 79.6  | 82.7  | (30) |  |
|                                                                        | 5%          | WB          | 71.5  | 70.4  | 69.4  | 67.7  | 62.3  | 56.6  | 55.8  | 59.1  | 62.1  | 65.3  | 68.8  | 70.6  | (31) |  |
|                                                                        |             | MCDB        | 79.1  | 78.8  | 77.6  | 74.3  | 70.7  | 66.9  | 65.7  | 71.0  | 75.4  | 78.0  | 78.1  | 80.7  | (32) |  |
|                                                                        | 10%         | WB          | 70.2  | 69.2  | 68.2  | 66.0  | 60.9  | 55.5  | 54.6  | 57.6  | 60.7  | 63.8  | 67.4  | 69.2  | (33) |  |
|                                                                        |             | MCDB        | 76.7  | 76.6  | 76.3  | 72.7  | 69.4  | 65.9  | 64.5  | 69.2  | 74.3  | 75.8  | 76.7  | 78.4  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 15.5  | 15.6  | 16.5  | 16.2  | 19.2  | 20.9  | 20.4  | 21.5  | 21.3  | 18.7  | 16.4  | 15.9  | (35) |  |
|                                                                        |             | MCDBR       | 19.9  | 19.4  | 21.1  | 21.2  | 23.4  | 25.8  | 26.3  | 28.4  | 29.0  | 27.8  | 24.6  | 22.6  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 7.1   | 6.8   | 7.3   | 9.3   | 9.3   | 10.2  | 10.2  | 10.8  | 9.8   | 8.7   | 8.7   | 8.7   | (37) |  |
|                                                                        |             | MCWBR       | 16.1  | 16.9  | 17.1  | 16.4  | 19.0  | 20.5  | 20.3  | 22.8  | 23.1  | 22.3  | 19.3  | 19.2  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.401 | 0.393 | 0.388 | 0.371 | 0.336 | 0.320 | 0.321 | 0.381 | 0.432 | 0.451 | 0.419 | 0.418 | (40) |  |
|                                                                        |             | taud        | 2.368 | 2.393 | 2.394 | 2.400 | 2.457 | 2.478 | 2.461 | 2.290 | 2.178 | 2.175 | 2.297 | 2.311 | (41) |  |
|                                                                        | Ebn at Noon |             | 299   | 297   | 289   | 279   | 276   | 275   | 279   | 271   | 271   | 276   | 292   | 295   | (42) |  |
|                                                                        |             | Edh at Noon | 42    | 40    | 38    | 35    | 30    | 28    | 29    | 38    | 46    | 49    | 45    | 44    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1914  | 1892  | 1677  | 1431  | 1322  | 1196  | 1269  | 1468  | 1658  | 1683  | 1802  | 1893  | (44) |  |
|                                                                        | RadStd      |             | 142   | 163   | 130   | 99    | 62    | 72    | 44    | 89    | 103   | 116   | 150   | 146   | (45) |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (8 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page